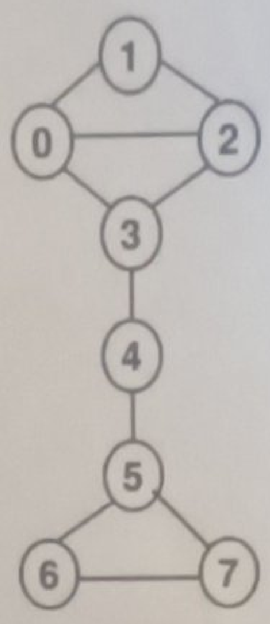
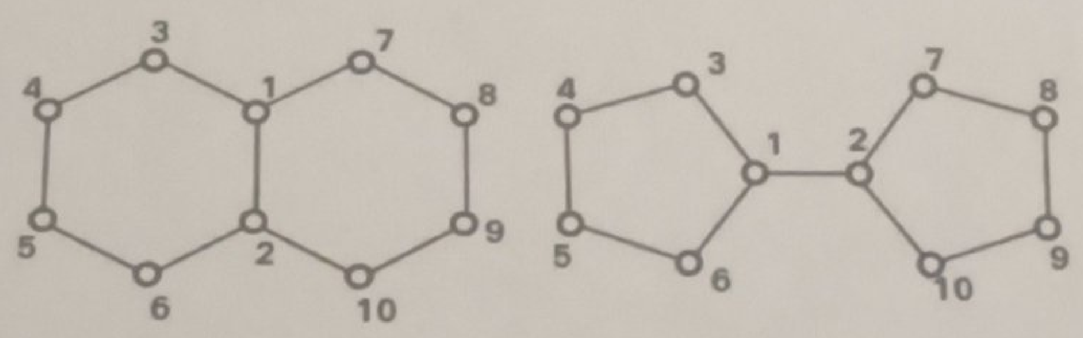
MidSem
Advanced ML
DSL606/DS603
Time 2 hrs

1. A. Given the following graph, find out the betweenness centralities of nodes - 0, and 1. What is the clustering coefficient of node 3 ? [2+2+2]



B. Consider a graph with three nodes, 1, 2, and 3 and three edges - (1,1) (self loop), (1,3), and (2,3). Find the Adamic-Adar and Katz Index for the missing link (1,2). For the Katz index, only compute paths upto length 2 and consider $\beta = 0.8$. [2+3]

C. Verify whether the following two graphs are equivalent using the color refinement algorithm (upto three iterations) with WL (Weisfeiler-Lehman) kernel. [4]

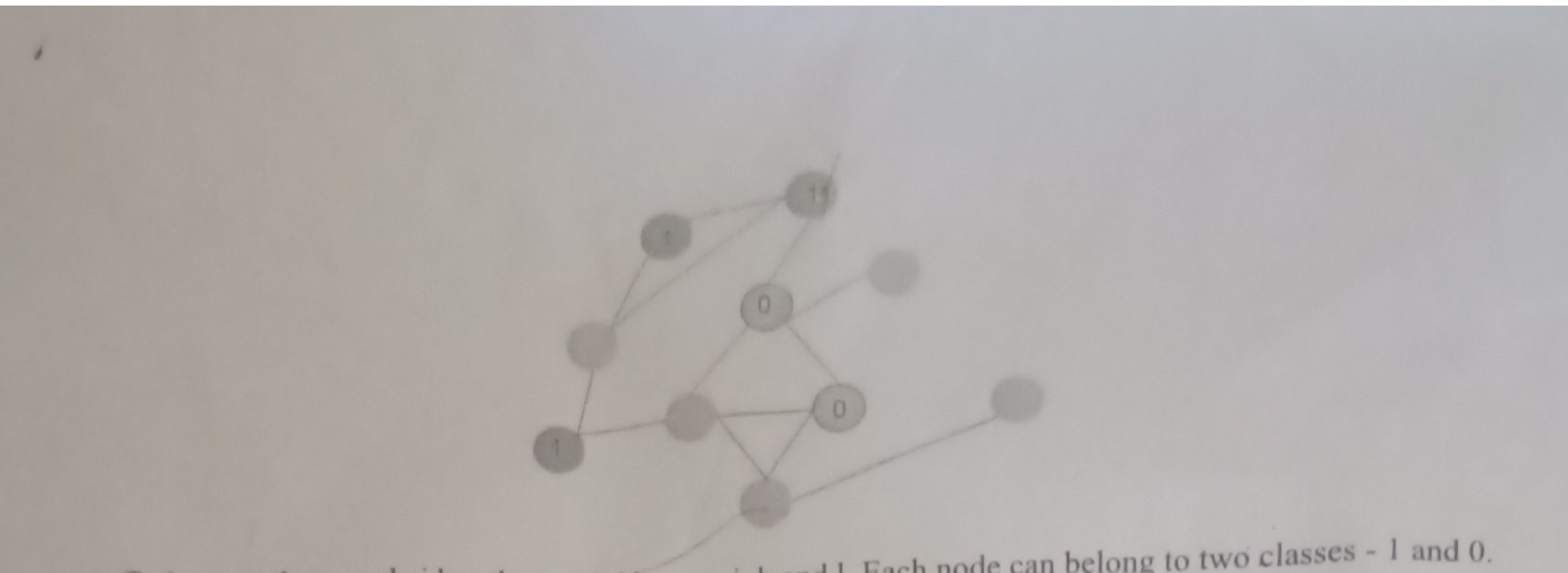


2. A. While finding node embeddings using random walk (say, for node u), one can minimize the following loss function -

$$\mathcal{L} = \sum_{u \in V} \sum_{v \in N_R(u)} -\log\left(\frac{\exp(\mathbf{z}_u^T \mathbf{z}_v)}{\sum_{n \in V} \exp(\mathbf{z}_u^T \mathbf{z}_n)}\right)$$

Where **z** represents the node embedding and $N_R(u)$ represents the random walk neighborhood of u. What is the time complexity of computing node embeddings in this manner? How can we improve it (explain with proposed modification in the equation)? [2+3]

B. In the following graph, use a probabilistic relational classifier and predict the labels of the unlabeled nodes given the labeled nodes in class 1 and class 0 (marked in figure). You need to run three iterations of the algorithm; after that mention which labels are converged and which are not. Further, irrespective of convergence, predict labels of all unknown nodes using threshold 0.5. [5]



C. Assume that a node i has three neighbours - j, k and l. Each node can belong to two classes - 1 and 0. Estimate the node i's message / estimate of node j being in class 1 following the loopy belief propagation algorithm. Given, label-label potential of class pair (1,1) is 0.7, (0,0) is 0.7 and (1,0) or (0,1) is 0.3; the prior belief of node i being in class 1 is 0.6 and class 0 is 0.4; k's message / estimate of i being in class 1 is 0.4, class 0 is 0.6; l's message/estimate of i being in class 1 is 0.5, class 0 is 0.5. [4]

3. A. Explain why CNN can be considered as a special case of GNN. [3]

B. In the following two scenarios, explain whether you would choose GCN, Relational GCN or GAT - i) Analyzing social networks where a node might be influenced differently by different individuals in its neighborhood; (ii) Modeling a knowledge graph where entities are connected by various relationships like "is located in," "is author of," and "is spouse of" etc. [2+2]

C. In the GRAFT-Net paper by Sun et al., how the authors use personalized PageRank while updating entity embeddings and why? [3]

4. A. Why does TransE Knowledge Graph Embedding not support 1-to-n relationships like 'teacher-student'; but TransR supports it? [3]

B. Why does TransE Knowledge Graph Embedding not support symmetric relationships like 'sibling'; but RotatE supports it? [3]

Indian Institute of Technology Bhilai

End-Semester Examination Winter 2025

Time: 3 Hr Adversarial Machine Learning (CSL607) Total: 100

Q1. In the AVIH paper, the concept of variance consistency loss is discussed. Write down the formula for the variance consistency loss as described in the paper. (Additionally, given the following 5×5 matrix M, calculate the variance consistency loss using a 3×3 kernel block:

(Marks: 20)

$$M = [[0.1, 0.2, 0.3, 0.4, 0.5],$$
$$[0.5, 0.4, 0.3, 0.2, 0.1],$$
$$[0.2, 0.3, 0.4, 0.5, 0.6],$$
$$[0.6, 0.5, 0.4, 0.3, 0.2],$$
$$[0.3, 0.4, 0.5, 0.6, 0.7]]$$

Q2. Describe Adversarial Attack Type 1 and Type 2 in the context of adversarial machine learning. What is the difference between the two attack types? (Marks: 10)

Q3. The StegaStamp decoder is trained using cross-entropy loss on the output message bits. Explain how the cross-entropy loss used here for recovering discrete binary data differs from the traditional cross-entropy loss used in classification. (Marks: 15)

Q4. The performance of Membership inference attacks can depend on training set composition. Discuss how identity-based selection versus random selection in GAN training datasets affects attack success. Why does identity-based selection increase vulnerability? (Marks: 10)

Q5. Propose and analyze a sophisticated adversarial strategy to remove or disrupt an embedded artificial fingerprint in a generated deepfake, assuming the adversary has significant computational resources but not necessarily the specific embedding key. (Marks: 10)

Q6. Describe the difference between black-box, white-box, and grey-box adversarial attacks. Why is defending against white-box attacks particularly difficult? (Marks: 10)

Q7. What is the transferability property of adversarial examples? Why do black-box attacks often succeed even when the attacker does not know the target model? (Marks: 10)

Q8. Suppose you have a most optimum generic adversarial (targeted) attack for a given classifier M and dataset $X \in R^{n,n}$ with labels $Y \in \{1, 2, ..., m\}$. Given such an attack, how will you perform adversarial training? Explain in the context of the min-max optimization problem.

(Marks: 15)

Advanced Machine Learning
DS606/DSL606
END SEM
3 hrs

1. (a) Give an example how model understanding can help in detecting potential bias in any model.
(b) How can the stability of post hoc explanations be measured mathematically?
(c) How scaffolding attacks can help in hiding the inherent bias present in the model?

[2.5+2+2.5]

2. Suppose we design an "Interpretable Decision Set" model (Lakkaraju et al) for a problem which gives the following four rules:

(i) If Respiratory-Illness=Yes and Smoker=Yes and Age≥ 50 then Lung Cancer - Satisfied by d1, d2
(ii) If Risk-LungCancer=Yes and Blood-Pressure≥ 0.3 then Lung Cancer - Satisfied by d2, d3
(iii) If Risk-Depression=Yes and Past-Depression=Yes then Depression - Satisfied by d1, d4, d5
(iv) If BMI≥ 0.3 and Insurance=None and Blood-Pressure≥ 0.2 then Depression - Satisfied by d2, d3, d5

Where d1, d2, …, d6 are six data points with d1, d2, d3 belonging to 'Lung Cancer' class and d4, d5, d6 belonging to 'Depression' class.

Show how you would measure the parsimony and distinctness of this set of rules for this dataset. Given the total number of input patterns is 10 and the maximum number of conditions which can be given on any input pattern is 4.

[2+4]

3. In "PRINCE: Provider-side Interpretability with Counterfactual Explanations in Recommender Systems" paper, how do the authors ensure counterfactuality and minimality of their explanations? No need to give correctness proof; only explain the relevant part of the method handling these two properties.

[3+2]

4. (a) In "ELIXIR: Learning from User Feedback on Explanations to Improve Recommender Models" paper, how the authors reduced the complexity of the search space for label propagation from O ($|I|^2$) to O ($|I|$)? [4]

(b) Suppose there are three items x,y and z with the following pairwise feedbacks - F(x,y)=1, F(y,z)=1 and F(x,z)=-1. The items are represented as x=[0.8], y=[0.5] and z=[0.3]. Compute the optimization function proposed in the paper, given, the current user preference vector is [0.1], user preference is applied as a translation and item-item similarities are calculated as cosine similarities. You can assume the regularization constant to be 0.1. [5]

5. Briefly mention the ideal scenarios for applying transfer learning, meta learning, multi task learning and contrastive learning. [One/two sentences per case] [4]

6. Consider the following scenario where you would like to use Model-Agnostic Meta Learning using a simple linear model: $f_\theta(x)=\theta x$. We have two meta training tasks (A & B) and there is only one datapoint

given for each task: (x=1,y=2) for task A and (x=1,y=−1) for task B. Show (a) one iteration of parameter update of each task and (b) one iteration of meta update. Assume, initial $\theta=0$; Learning rate for task-specific updates for both task A and B: $\alpha=0.5$; Meta learning rate: $\beta=0.1$; and use Mean Squared Error (MSE) loss: $L=1/2(f_\theta(x)-y)^2$ for all cases. [2+2+3]

7. a) Let us assume a federated learning setup with two clients both using linear models: $f_\theta(x)=\theta x$. Each client has one datapoint: (x=-1,y=2) for client 1 and (x=1,y=0) for client 2. Assume initial global $\theta=0.0$, Mean Squared Error (MSE) loss: $L=1/2(f_\theta(x)-y)^2$ and learning rate=0.1 for both clients. Show one iteration of calculation in the clients and the corresponding updated parameter value at server using fed avg. [2+2+1]

b) Now, assume we want to ensure local differential privacy in this framework with gradient clip at 1.0 and noise 0.05 (assuming constant noise for simplicity). Show one iteration of calculation in the clients and the corresponding updated parameter value at server using fed avg. in this locally differentially private setup. [3]

8. In "Meta-Exploiting Frequency Prior for Cross-Domain Few-Shot Learning" by Zhou et al., briefly explain how they have incorporated 'Frequency prior regularization' (show equation) and the rationale behind using it. [2+2]

9. In "First-Explore, then Exploit: Meta-Learning to Solve Hard Exploration-Exploitation Trade-Offs" paper, which limitation of the Meta-RL literature is targeted to be solved? Why do the authors use two different policies here? How are these policies dependent on each other? [2+2+2]

Mid Semester Examination
Introduction to Formal Verification and Its Application CS511
2 Hours, Maximum Marks 80

- **ALL PARTS of a question MUST be answered in sequence**
- **Neat drawing is mandatory. Striking off and cluttering of drawings may lead to non-evaluation/non-marking of such answers.**
- **If any answer is expected to explained with an "example" no credit will be awarded for only theory**

Question 1 ----------[7.5 + 10 + 7.5]

Consider the SOP: $f(w,x,y,z) = x'y' + wxy + x'yz'$.

- Step wise show minimization using Tabular Method. Discuss the complexity of the scheme.
- Now, show the minimization using selection of Prime Implicants and unate covering scheme. Discuss the complexity of this scheme.
- Step wise show minimization using Espresso Algorithm.

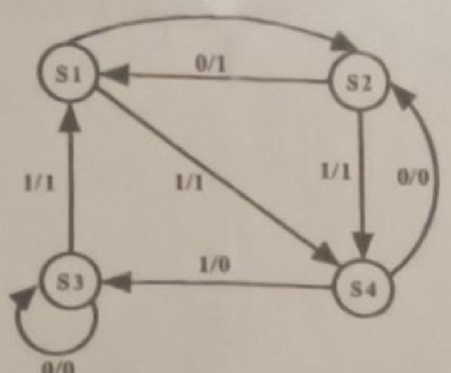
Finally draw a comparison of the three methods.

Note:

- Step wise illustration is mandatory. Skipping of steps will lead to deduction of marks.
- Define all terms; for example, Prime Implicants, essential prime Implicants etc.

Question 2: [Marks 15]

Consider the following FSM



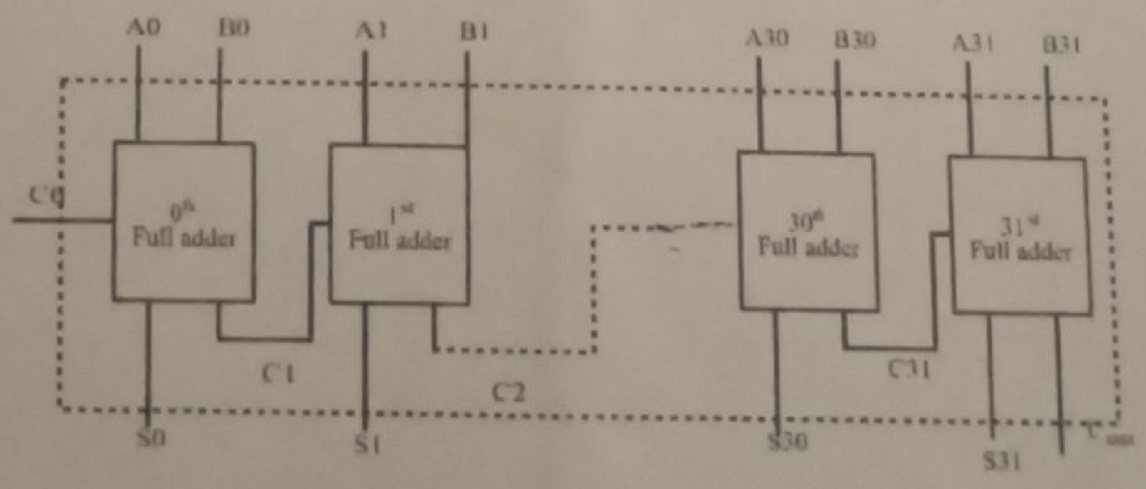
Optimize the FSM in terms of state minimization by merging equivalent states and optimal state encoding. **Note**: Step wise illustration is mandatory. Skipping of steps will lead to deduction of marks.

Question 3: [Marks 20]

Map the scheduling problem to 0-1 Integer Linear Programming. What does it show about the complexity of the scheduling problem? Consider any heuristic for the high-level synthesis problem. Prove using an example (at-least 4-time steps and 10 operations) that the heuristic provides a sub-optimal solution compared to ILP.

Question 4: [Marks 20]

Consider a 32 bit full adder. It needs to be tested for proper functoriality. Propose a Design for Testability for testing of the circuit at reasonable time required for testing.



DS209: Information Security **End-sem exam.**

Maximum marks: 150

Maximum duration: 0930 – 1200 hrs (150 mins) *Date: December 08, 2023*

Instruction. Answer clearly and to the point. Show steps for getting credits for partial answers.

Question 1 *Answer exercises 1, 2 and 3 of the attached sheet.* *(15+15+10=40)*

Question 2 *Describe the following pieces of malware: virus, worm, logic bomb, trojan horse, rootkit, keylogger, spyware, adware, bot and mobile code.* $(10 \times 2 = 20)$

Question 3 *(A) Describe the usefulness of all the bits in various permission triads in a Linux file system. (B) How a user is able to change her own password in the Linux file system, but is unable to change the password of others. (C) Clearly describe a TOCTTOU attack with diagrams. (D) Clearly describe how the setuid bit is used to securely printing files by various users.* $(5 + 10 \times 2 + 5 = 30)$

Question 4 *Define the fairness property of a protocol. Design a 2-party lottery protocol which is fair. Give all details of the programs (a.k.a. smart-contracts) in a pseudocode format. Explain in detail why the protocol is fair.* (30)

Question 5 *With clear and precise diagrams, show how exactly a buffer-overflow attack happens (as described in the shared video lecture). Show all the steps with elaborate explanation.* *(30)*

Zero-knowledge proof:

- To prove that you know/have something, without revealing the knowledge.
- Example: $Z_p^* = \{1, 2, 3, \ldots, p-1\}$. p is a large prime.

  $Z_p^* = \{g, g^2, g^3, \ldots, g^{p-1}\} = \langle g \rangle$

  — Z_p^* is a cyclic group with generator g.

  — Concrete example: $p = 11$

  $g = 2$

$$Z_{11}^* = \{\underset{2}{2}, \underset{4}{2^2}, \underset{8}{2^3}, \underset{5}{2^4}, \underset{10}{2^5}, \underset{9}{2^6}, \underset{7}{2^7}, \underset{3}{2^8}, \underset{6}{2^9}, \underset{1}{2^{10}}\}$$ [all operations are mod p]

Suppose, $y \longleftarrow \$ Z_p^*$ and g is known.
Can you compute $x \in Z_p^*$ such that $g^x = y$?
This is a hard problem, if p is large.
This problem is called discrete logarithm problem.

Suppose, Alice knows x and $g^x = y$. Bob knows $g^x = h$.
How can Alice prove to Bob that she knows x.
A simple protocol is below.

A (x, h, g) — B (h, g)

A → B: x

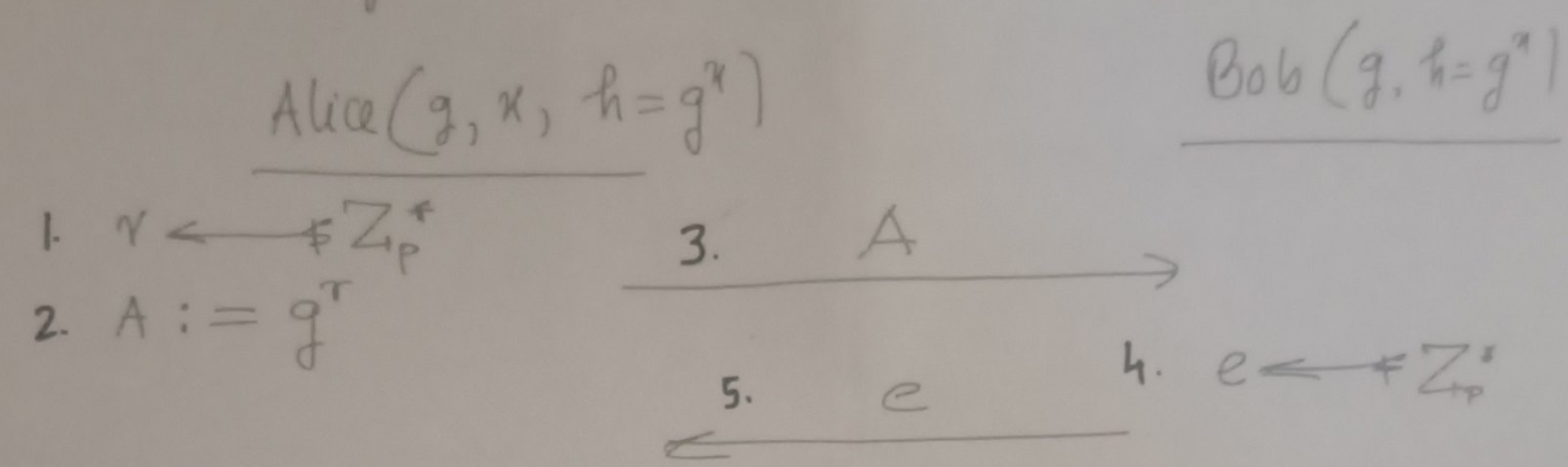
B checks: $g^x \stackrel{?}{=} h$

- The main issue with this protocol is that Bob now knows the secret.
- Is it possible to design a protocol s.t.

Alice can prove to Bob "knowledge" of x without revealing x.

This is The solution is not easy

- Schnorr has devised a protocol (which year?)
- Following is the Schnorr algorithm.



Convince yourself then

✓1. If Alice knows x, the checking gives 1. (correctness property)

✓2. If Alice does not know x, then she cannot "somehow" generate A and δ, so that the checking gives 1. (Soundness property)

✓3. B cannot extract x from A and δ. (Zero-knowledge property)

Exercise!

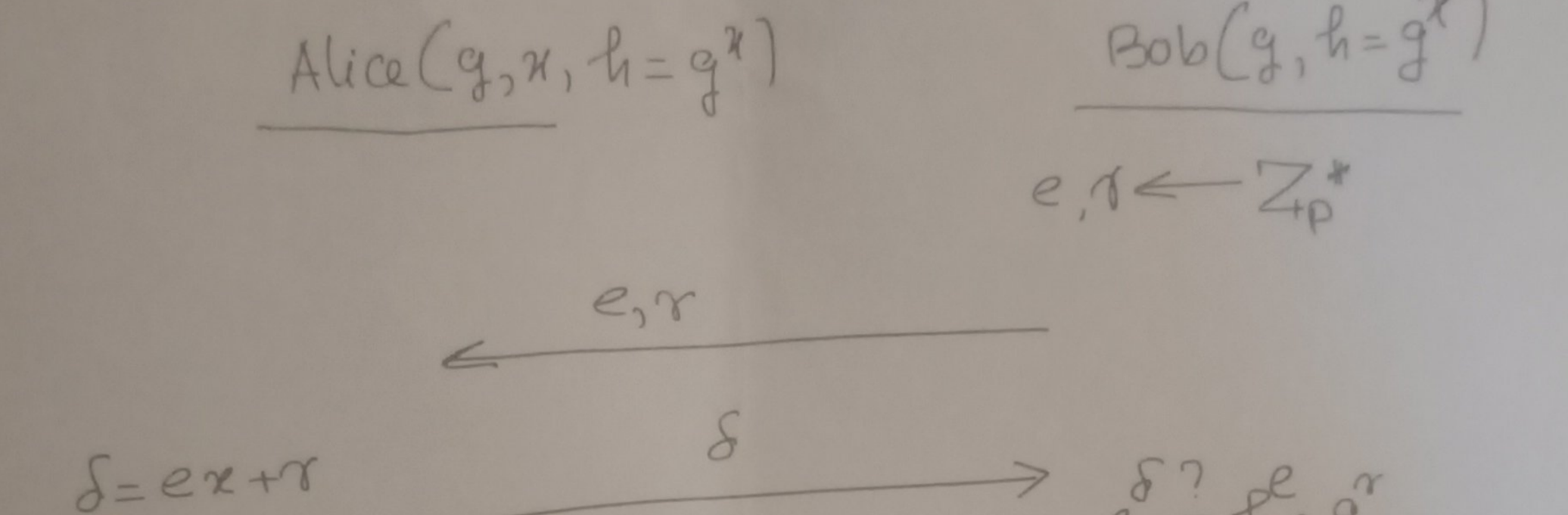
What happens if the above protocol happens in the following order:

4, 5, 1, 2, 3, 6, 7, 8

- Is the correctness property satisfied?
- Is the soundness property satisfied?
- Is the zero-knowledge property satisfied?

Exercise 2

Are the correctness, soundness and zero-knowledge properties of the following protocol satisfied?



EXERCISE 3

Coming back to Schnorr's original ~~proof~~ zero-knowledge proof protocol for (x, g, g^x). You can see that the protocol is inherently interactive. In fact there are 3 rounds of communication. Can we design a modified version of Schnorr's protocol that is one-round?

Artificial Intelligence : DS251/CSL304

MidSem

Time - 120 Minutes

1. The classic Tower of Hanoi puzzle consists of three poles and a set of n disks of different sizes which can slide onto any pole. The puzzle starts with the disks stacked in ascending order of size on one pole (the source pole), with the smallest disk on top. The objective is to move the entire stack to another pole (the target pole) with the help of the third pole (the helper pole), following these rules: (i) Only one disk can be moved at a time, (ii) A disk can only be moved if it is the top disk on a pole, (iii) No disk may be placed on top of a smaller disk.

Formulate this puzzle as an AI search problem by defining:

a. State representation: What does each state represent?

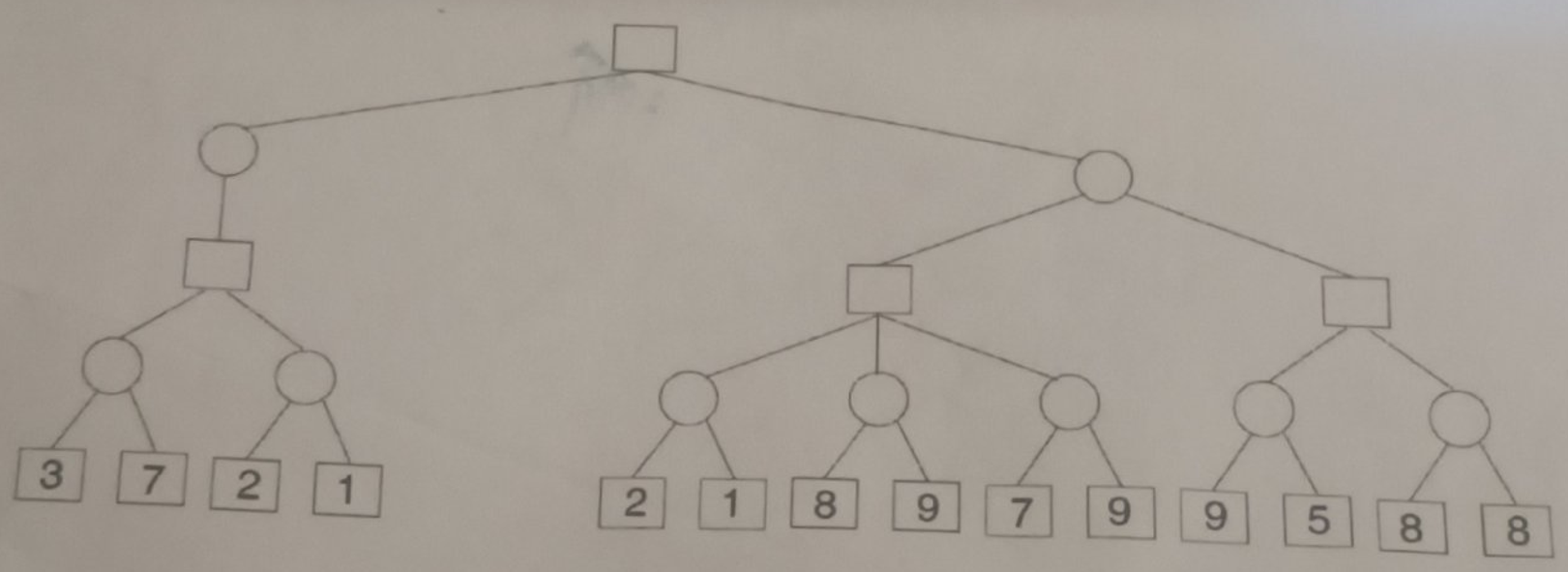
b. Initial state and Goal State.

c. Actions: Define the possible moves from one state to another and corresponding costs.

Draw the search tree for the first two levels of the search space starting from the initial state.

(2+1+2+2)

2. Apply α-□ pruning on the following tree. Show the (α, □) value of each node and indicate which branches to prune and why. The rectangles indicate "Max" nodes and circles indicate "Min" nodes. *(5)*



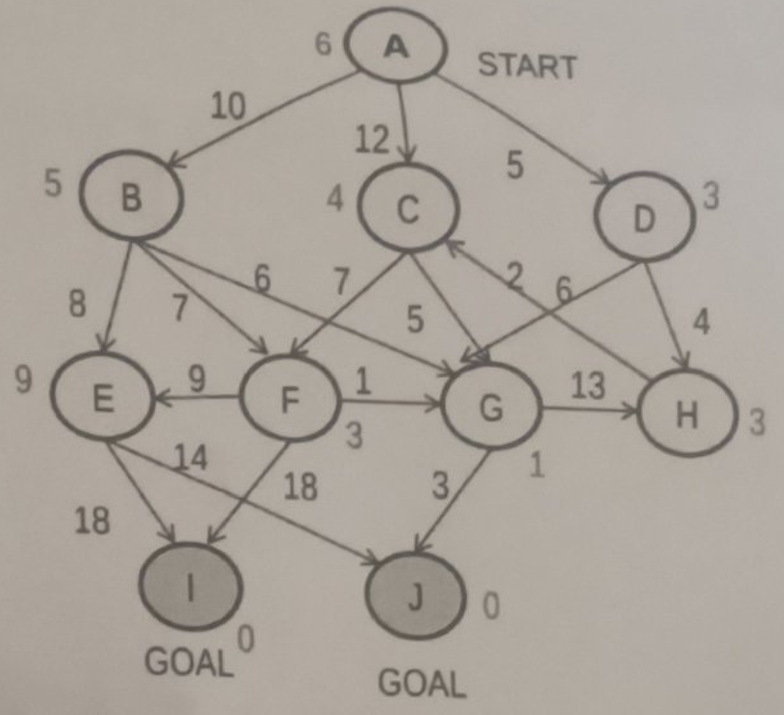
3. Let's say, you are going to spend a month in the wilderness. The only thing you are carrying is the backpack which can hold a maximum weight of 40 kg. Now you have different survival items, each having its own "Survival Points" (which are given for each item in the table). Some of the items are so essential that if you do not take them, you incur some additional penalty.

Here is the table giving details about each item.

Item	Weight	Survival Value	Penalty if not taken
Sleeping Bag	30	20	0
Rope	10	10	0
Bottle	5	20	0
Torch+Battery	15	25	-20
Glucose	5	30	0
Pocket Knife	10	15	-10
Umbrella	20	10	0

Formulate this as a genetic algorithm problem where your objective is to maximize the survival points. Write how you would represent the chromosomes, fitness function, crossover and mutation. (3+2+2+1)

4. The following is a representation of a search problem. There is a heuristics h which is marked beside every node whereas the path costs are marked beside every edge. Run A* on this search space and show the final path with steps. (6)



Clarification: h(A)=6, h(B)=5, h(C)=4, h(D)=3, h(E)=9, h(F)=3, h(G)=1, h(H)=3, h(I)=h(J)=0

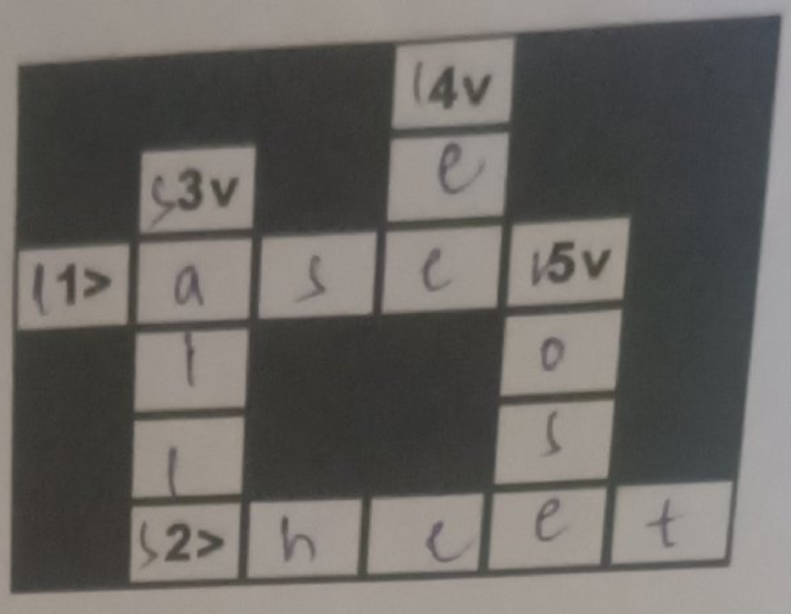
5. Consider the following axioms:

a. All hounds howl at night.
b. Anyone who has any cats will not have any mice.
c. Light sleepers do not have anything which howls at night.
d. John has either a cat or a hound.
e. (Conclusion) If John is a light sleeper, then John does not have any mice.

Prove the conclusion using the resolution refutation method. Show all required steps like conversion to CNF, skolemization etc. (10)

Consider the following crossword puzzle. The arrow direction shows whether it needs to be filled horizontally and vertically from that position - for instance five letters horizontally from position 1, four letters vertically from position 5 etc. All positions should have distinct words. From each position, you need to fill up the maximum number of positions in the given direction i.e. from position 1 you have to fill up five letters horizontally, not three or four letters.

Available words: {aft, ale, eel, force, heel, hike, keel, knot, laser, lee, line, rose, roses, sails, sheet, steer, tie}



a. Define the variables and their domains.
b. Draw the constraint graph.
c. Run ARC consistency (show steps) and after that the updated domains of each variable.
d. Select the first variable to assign and which value to assign.

(2+2+4+2)

INDIAN INSTITUTE OF TECHNOLOGY BHILAI

CSL 504: Graph Theory(2025-26M)

End Semester Examination

Marks 40 Time 180 Minutes

---

1. Let $G = (V, E)$ and $|V| = n$. Suppose $x, y \in V$, $e = (x, y) \in E$ and $degree(x) + degree(y) \geq n$. Then $G + e$ is Hamiltonian iff G is Hamiltonian. [10]

2. Let G be an $s - t$ network with a flow f. Prove the following statements:

   (a) Let (A, B) be any $s - t$ cut. Then $value(f) \leq capacity(A, B)$. [4]

   (b) If f is a maximum flow, then there exist an $s - t$ cut (A', B') such that $value(f) = capacity(A', B')$. [5]

3. Prove that every strongly connected tournament has a hamiltonian cycle. [8]

4. Prove that $R(3, 3) = 6$. [3]

5. (a) Define k-critical graph. [2]

   (b) Show that if G is k-critical, then $\delta \geq k - 1$. [3]

6. Consider the following network and find the maximum flow in the network using Ford-Fulkerson Algorithm.

   (a) In each step of the algoirthm, you must choose the longest path (that involves maximum number of edges) to send the flow. Draw the residual graph in each step. [5]

   (b) Show the cut in the network that has minimum capacity. [3]

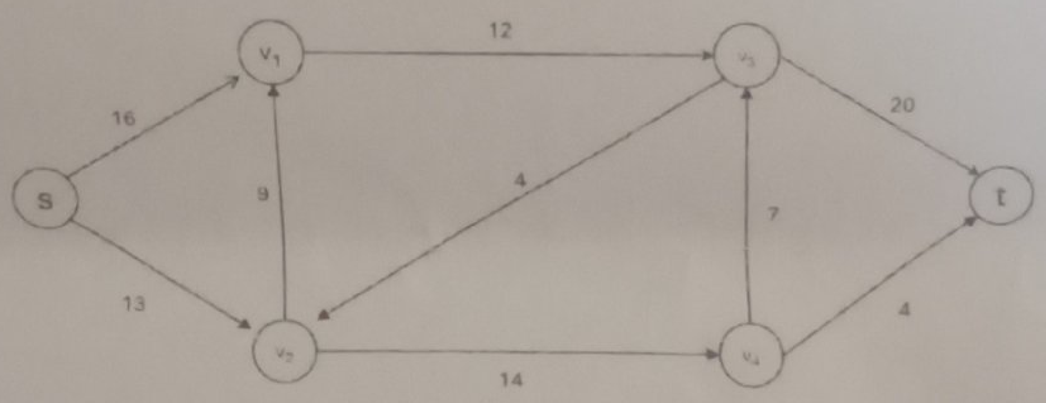


Figure 1: Example Graph for Question 6

INDIAN INSTITUTE OF TECHNOLOGY BHILAI

CSL 504: Graph Theory(2025-26M)

Mid Semester Examination

Marks 25 Time 120 Minutes

---

1. Let G be an undirected connected graph. Show that number of odd degree vertices in G is always even. [2]

2. Show that if the diameter of a simple graph G is at least four, then the diameter of its complement $\overline{G}$ (that includes only those edges which are not in G) is no more than two. [3]

3. Prove that either a graph G or its complement $\overline{G}$ (that includes only those edges which are not in G) is connected. [3]

4. Let M and M' be two matchings in an X,Y-bipartite graph. Suppose that M saturates $S \subseteq X$ and M' saturates $T \subseteq Y$. Prove that G has a matching that saturates $S \bigcup T$. [5]

5. Let G be a graph without isolated vertices, then show that $\alpha'(G) + \beta'(G) = n(G)$, where $\alpha'(G)$ and $\beta'(G)$ denotes the maximum matching and minimum size edge cover in G, respectively. [4]

6. Show that if G is 3-regular, then $\kappa(G) = \kappa'(G)$. [3]

7. Give an example of a graph where $\kappa < \kappa' < \delta$ [2]

8. Define Block of a graph. Show that two blocks of a graph can share at most one vertex. [3]

Mid Semester Examination, CAD for VLSI CSL605/CS558
3 Hours, Maximum Marks 100

- If you are asked to answer a question using example, No marks will be awarded only for theoretical explanations.
- Draw neat Circuit diagrams. Overwriting/Striking off etc., if done on circuit diagrams, they will not be checked

---

1. Explain using examples the philosophy "VLSI Testing is more complex compared to classical system testing" ----------**5 Marks**
2. It is said that "If a circuit is tested to have a stuck at 0/1 fault at a line X say, it cannot be guaranteed that really net X is stuck at ground/power". Of course you can say that circuit is not normal.
   a. Still stuck at fault model is widely accepted. Explain why is it so? -----**5 Marks**
   b. Take an example of a circuit without fanouts. Show that if s-a faults are considered only at the primary input lines we get 100% coverage. What will be the case if the circuit has fanouts? ----------**10 Marks**
   (Explain all intermediate steps used like fault collapsing, faults at fan-out lines etc.)

3. Sensitize-propagate-justify (D algorithm) guarantees to generate a test pattern for a fault if it is testable, while random pattern-based test pattern generation schemes may sometimes take long time for some faults. Still for about 90% of the faults we use random pattern-based test pattern generation schemes while for remaining 10% of faults we use D algorithm. Also explain how to determine easy and difficult to test faults with example.
   Also explain how to determine easy and difficult to test faults with example and SCOAP.

   Explain the concept using an example of a circuit from the context of time complexity, space complexity etc. You may assume that "parallel fault simulation" is used as fault simulator for random pattern based test pattern generation scheme.----------**25 Marks**

4. In a sequential circuit if there are flip-flops up to depth "n" then use of time frame expansion method for test pattern generation is broadly $O(n.k)$, where k is the order of complexity for test pattern generation by a combinational version of the sequential circuit. The combinational version of the sequential circuit is created by removing all flip-flops and shorting input "D" with output "Q". Illustrate this using an example. (Explain all intermediate steps)----------**20 Marks**

5. Take a circuit with 3 inputs and 1 output, with 3 levels. Design the BIST circuit with LFSR and Response compaction. Illustrate testing of two different faults and the concept of fault aliasing.----------**25 Marks**

6. Explain using an example the following fact "March test can detect all the rising transient faults in a memory" ----------**10 Marks**

Artificial Intelligence : DS251/CSL304

EndSem

Time - 3 hrs

Full Marks 65

1. The monkey-and-bananas problem is faced by a monkey in a laboratory with some bananas hanging out of reach from the ceiling. A box is available that will enable the monkey to reach the bananas if he climbs on it. Initially, the monkey is at A, the bananas at B, and the box at C. The monkey and box have height Low, but if the monkey climbs onto the box he will have height High, the same as the bananas. The actions available to the monkey include Go from one place to another, Push an object from one place to another, ClimbUp onto or ClimbDown from an object, and Grasp or Ungrasp an object. The result of a Grasp is that the monkey holds the object if the monkey and object are in the same place at the same height. The goal here is to make the money grasp the bananas hanging from the ceiling.

a. Write down the initial state description.
b. Write the six action schemas.
c. Apply the partial-order planning algorithm to draw the final partial order plan clearly showing the causal links (including the subgoals that they achieve) and ordering links.

[2+3+7=12 Marks]

2. a. Consider the following Car Tire replacement planning problem. Draw the GraphPlan for the first layer and show all the mutex links.

Start: At(Flat, Axle) ∧ At(Spare, Trunk)
Goal: At(Spare, Axle)

Op(ACTION: Remove(Spare, Trunk), PRECOND: At(Spare, Trunk), EFFECT: At(Spare, Ground) ∧ ¬ At(Spare, Trunk))

Op(ACTION: Remove(Flat, Axle), PRECOND: At(Flat, Axle), EFFECT: At(Flat, Ground) ∧ ¬ At(Flat, Axle))

Op(ACTION: PutOn(Spare, Axle), PRECOND: At(Spare, Ground) ∧ ¬ At(Flat, Axle), EFFECT: At(Spare, Axle) ∧ ¬ At(Spare, Ground))

Op(ACTION: LeaveOvernight, PRECOND: None, EFFECT: ¬ At(Spare, Ground) ∧ ¬ At(Spare, Axle) ∧ ¬ At(Spare, Trunk) ∧ ¬ At(Flat, Ground) ∧ ¬ At(Flat, Axle))

b. *In GraphPlan, explain whether the following statements are true or false with justifications - (i) Literals increase monotonically; (ii) Actions increase monotonically; (iii) Mutexes increase monotonically.*

[7+3=10 Marks]

3. Consider the following axioms,

If the alarm sounds, then there is a fire or there is a drill.
If there is a drill, then the fire department will not come.
The fire department comes.
Therefore, if the alarm sounds, then there is a fire.

Represent all of them in Propositional Logic. Prove the conclusion using the resolution refutation method. Show all required steps.

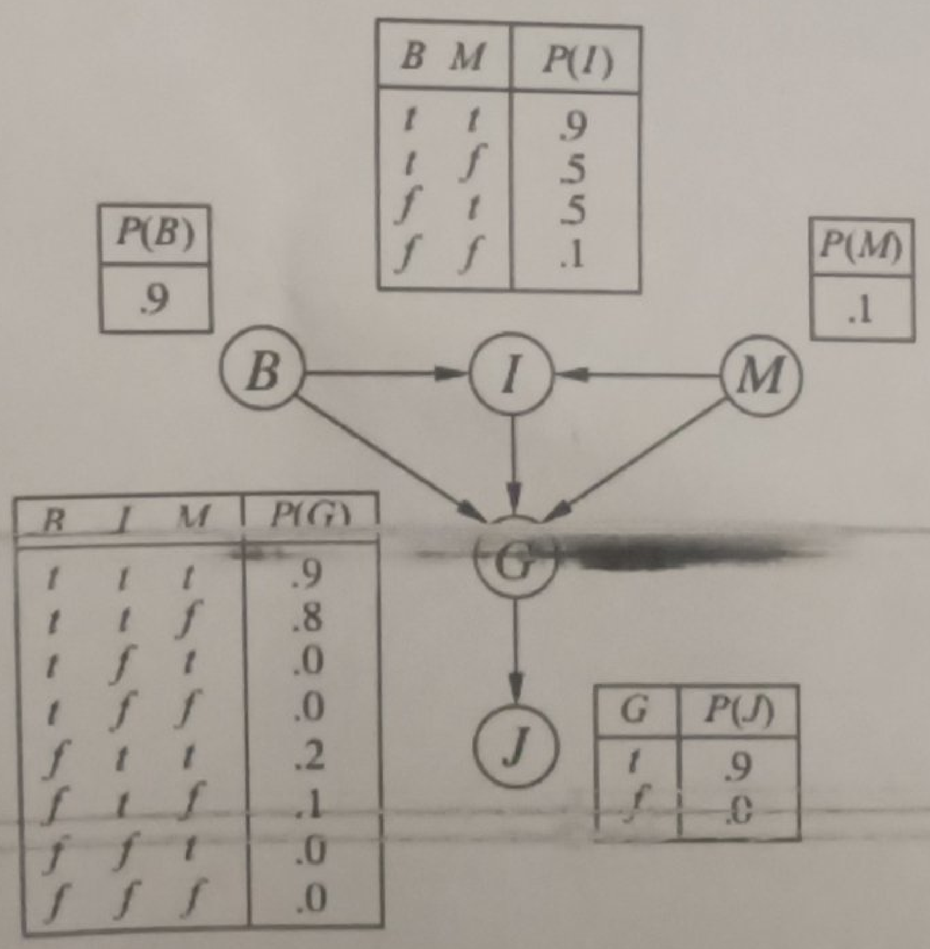
[6 Marks]

4. a. Consider two medical tests, A and B, for a virus. Test A is 95% effective at recognizing the virus when it is present, but has a 10% false positive rate (indicating that the virus is present, when it is not). Test B is 90% effective at recognizing the virus, but has a 5% false positive rate. The two tests use independent methods of identifying the virus. The virus is carried by 1% of all people. Say that a person is tested for the virus using only one of the tests, and that test comes back positive for carrying the virus. Which test returning positive is more indicative of someone really carrying the virus? Justify your answer mathematically.

b. Let's consider two fuzzy sets A and B defined on the same universe X={1,2,3,4,5} where the corresponding membership functions defined as
μ_A={(1,0.2),(2,0.5),(3,0.7),(4,0.4),(5,0.1)} and μ_B={(1,0.6),(2,0.3),(3,0.4),(4,0.8),(5,0.2)}.
Find $\mu_{A\cap B}(x)$ and $\mu_{AUB}(x)$.

[5+3 = 8 Marks]

5. Consider the following bayesian belief network,



Where boolean variables B indicates "BrokeElectionLaw" , I indicates "Indicted (charged for a crime)", M indicates "PoliticallyMotivatedProsecutor" , G indicates "FoundGuilty", and J indicates "Jailed" .

a. Which of the following are asserted by the network structure? Justify your answer.

(i) P(B, I, M) = P(B)P(I)P(M).

(ii) P(J | G) = P(J | G, I).

(iii) P(M | G, B, I) = P(M | G, B, I, J).

b. Calculate the value of P (B, I, ¬M, G, J).

c. Calculate the probability that someone goes to jail given that they broke the law, have been indicted, and faced a politically motivated prosecutor.

d. A context-specific independence allows a variable to be independent of some of its parents given certain values of others. In addition to the usual conditional independences given by the graph structure, what context-specific independences exist in the Bayes net in the Figure above?

[3+2+3+2=10 Marks]

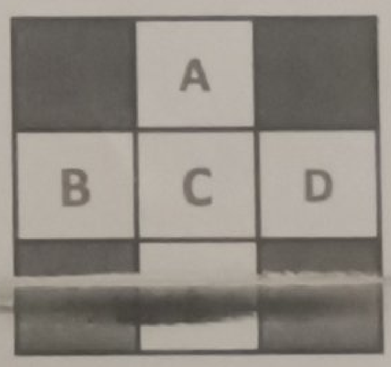
6. Consider an undiscounted MDP having three states, (1, 2, 3), with rewards −1, −2, 0, (can also be considered as initial values of the states) respectively. State 3 is a terminal state. In states 1 and 2 there are two possible actions: a and b. The transition model is as follows:

a. In state 1, action a moves the agent to state 2 with probability 0.8 and makes the agent stay put with probability 0.2.
b. In state 2, action a moves the agent to state 1 with probability 0.8 and makes the agent stay put with probability 0.2.
c. In either state 1 or state 2, action b moves the agent to state 3 with probability 0.1 and makes the agent stay put with probability 0.9.

Apply policy iteration, showing each step in full, to determine the optimal policy and the values of states 1 and 2. Assume that the initial policy has action b in both states.

[8 Marks]

7. Consider the following Gridworld example. We would like to use TD learning to find the values of these states.



Suppose we observe the following (s, a, s', R(s, a, s')) transitions and rewards:
(B, East, C, 2), (C, South, E, 4), (E, North, A, 4), (C, East, A, 6), (B, East, C, 2)

Note that the R(s, a, s') in this notation refers to observed reward, not a reward value computed from a reward function. The initial value of each state is 0. Let, discount factor $\gamma = 1$ and learning rate $\alpha = 0.5$.

What are the learned values for each state from TD learning after all five observations/samples?

[4 Marks]

8. Consider a Markov Decision Process (MDP) with two states $S=\{s_1,s_2\}$ and two actions $A=\{a_1,a_2\}$. The discount factor is $\gamma=0.8$. You are given the following information for a single transition:

- Current state: $s_t=s_1$
- Action taken: $a_t=a_1$
- Reward received: $r_t=5$
- Next state: $s_{t+1}=s_2$
- Current Q-values: $Q(s_1,a_1)=2$, $Q(s_2,a_1)=4$, $Q(s_2,a_2)=1$
- Learning rate: $\alpha=0.5$

Compute the updated value of $Q(s_1,a_1)$ after the transition.

b. Give an example situation where we cannot use traditional tabular Q-Learning and have to use Deep Q Learning.

c. Explain how exploration functions stop exploring actions whose badness is established.

[3+2+2=7 Marks]